

Problem Set 4

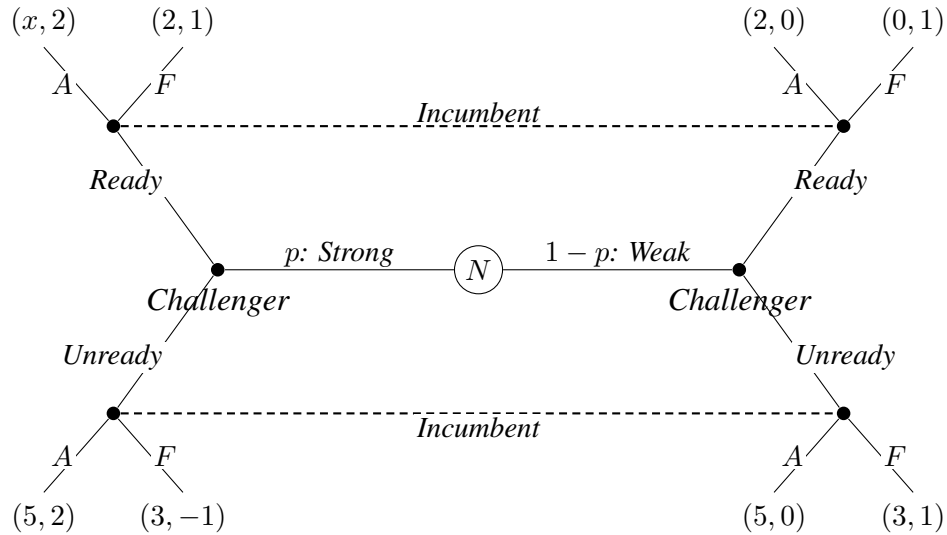
POLS W4732

Fall 2026

Due November 6 before the TA section

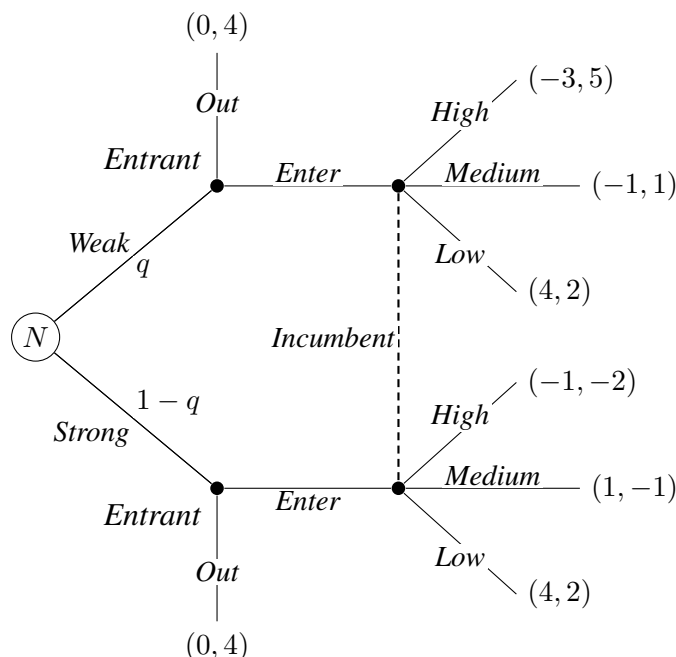
Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1 (Crisis bargaining). *Nature draws a Strong Challenger with probability p and a Weak Challenger with probability $1-p$. The Challenger observes its type and chooses Ready or Unready. The Incumbent observes that action but not the type and chooses A or F. Payoffs are shown as (Challenger, Incumbent).*



The Strong Challenger's payoff at (Strong, Ready, A) is the parameter x . For each candidate separating profile, derive both types' incentive constraints, the Incumbent's sequential-rationality conditions, and Bayes-consistent beliefs. Characterize all x for which a separating PBE exists.

Exercise 2 (Informed entry). *Nature draws a Weak entrant with probability q and a Strong entrant with probability $1-q$. The entrant observes its type and chooses Out or Enter. After Enter, Incumbent cannot observe type and chooses High, Medium, or Low.*



After supplying all required on- and off-path beliefs, classify each strategy profile below as (i) impossible (to support as a PBE) for every belief, (ii) possible for some off-path belief, or (iii) possible only for a unique belief.

1. Both types choose Out; Player 2 chooses High after Enter.
2. Both types choose Enter; Player 2 chooses Low.
3. Strong chooses Enter; Weak chooses Out; Player 2 chooses Low.

Then characterize, as a function of q , when either of two types of pooling strategy profiles can be supported as a PBE and how Player 2's action changes at the threshold.

Exercise 3 (Spence signaling). There are two types of worker. Type $\theta \in \{1, 2\}$ has productivity θ . The cost of sending signal z for the low type is $C_1(z) = z$ and for the high type is $C_2(z) = z(1 - \varepsilon)$. With probability π , the worker is a low type. The wage is assumed to equal the worker's expected productivity.

1. Show that there is a continuum of PBEs in which Type 1 chooses $z = 0$ and Type 2 chooses $z \in Z \equiv (1, \frac{1}{1-\varepsilon})$.
2. Consider now the possibility of a pooling equilibrium. What are the possible values of z compatible with a pooling equilibrium?
3. What is the highest equilibrium payoff to each type of worker from a pooling equilibrium? Can we conclude that Type 2 is always better off under a separating equilibrium?